

VICHY-CHARMEIL AP, France

WMO: 073740

Lat: 46.1667N

Lon: 3.3986E

Elev: 249

StdP: 98.37

Time Zone: 1.00 (EUC)

Period: 99-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-7.6	-5.2	-10.2	1.6	-6.2	-7.5	2.0	-3.5	8.5	7.0	7.5	6.5	1.0	190	0.469

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.9	33.5	20.7	31.1	20.0	29.0	19.3	21.7	30.2	20.8	28.8	20.0	27.3	2.4	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.2	14.4	24.5	18.3	13.6	23.3	17.5	12.9	22.4	64.3	30.3	61.1	28.8	58.3	27.4	25.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
7.2	6.2	5.3	-10.6	36.9	3.3	2.4	-13.0	38.6	-15.0	40.0	-16.8	41.3	-19.3	43.0		
			-10.9	23.6	3.3	1.0	-13.3	24.4	-15.3	24.9	-17.1	25.5	-19.5	26.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	11.6	3.7	4.2	7.5	10.4	14.3	18.4	20.2	19.8	16.2	12.7	7.4	4.4
	DBStd	7.04	4.28	4.38	3.72	3.42	3.52	3.67	3.35	3.42	3.29	4.08	4.02	4.24
	HDD10.0	799	200	166	94	36	4	0	0	0	1	21	97	179
	HDD18.3	2661	454	396	337	238	132	44	18	22	79	179	328	433
	CDD10.0	1392	3	3	16	49	139	251	315	303	188	104	19	4
	CDD18.3	214	0	0	0	0	8	46	75	67	16	3	0	0
	CDH23.3	2659	0	0	1	23	138	576	888	781	218	34	1	0
	CDH26.7	1029	0	0	0	2	32	225	368	344	56	3	0	0
Wind	WSAvg	2.0	2.1	2.4	2.4	2.3	2.0	1.9	1.9	1.7	1.6	1.7	1.9	2.1
Precipitation	PrecAvg	784	49	41	42	67	83	75	79	82	71	72	73	51
	PrecMax	963	102	87	108	150	151	186	184	192	149	147	139	126
	PrecMin	543	12	13	10	11	23	27	19	25	23	19	34	14
	PrecStd	111	22	19	21	36	34	37	35	41	32	32	25	27
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	15.5	18.0	22.0	26.2	29.9	35.6	35.9	37.6	31.4	26.7	20.1	16.4
		MCWB	11.2	10.6	13.0	15.9	19.0	21.0	20.4	20.6	18.7	17.6	14.2	11.7
	2%	DB	13.6	15.1	19.3	23.2	27.0	31.8	33.0	33.5	28.1	24.0	17.7	14.3
		MCWB	10.3	9.7	11.6	14.4	18.0	19.9	20.8	20.6	18.4	17.0	13.0	10.6
	5%	DB	12.0	13.0	17.1	20.6	24.4	29.2	30.7	30.6	25.9	21.7	15.7	12.8
		MCWB	8.9	8.6	10.8	13.3	16.6	19.6	20.2	20.0	17.7	16.0	12.1	9.7
	10%	DB	10.2	11.2	14.9	18.1	22.1	26.7	28.4	27.8	23.6	19.6	13.8	11.0
		MCWB	7.7	7.8	9.9	12.0	15.7	18.8	19.3	19.2	17.1	15.0	11.0	8.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.0	11.3	13.7	16.8	20.3	23.0	22.8	22.5	20.7	18.8	15.0	12.2
		MCDB	14.9	16.0	19.4	24.1	27.4	31.1	31.3	32.1	26.8	24.3	18.8	15.1
	2%	WB	10.6	10.1	12.3	15.0	18.7	21.4	21.6	21.4	19.5	17.5	13.6	11.1
		MCDB	13.2	13.8	17.7	21.6	25.0	29.3	30.2	30.2	25.5	22.4	16.6	13.7
	5%	WB	9.2	9.1	11.3	13.7	17.4	20.3	20.7	20.5	18.4	16.4	12.5	10.0
		MCDB	11.5	12.6	16.0	19.2	22.7	27.3	28.9	28.5	24.2	20.6	15.3	12.4
	10%	WB	8.0	8.1	10.3	12.4	16.1	19.2	19.8	19.6	17.4	15.4	11.2	8.7
		MCDB	10.0	10.9	14.4	17.3	20.9	25.4	27.0	26.6	22.6	19.0	13.5	10.9
Mean Daily Temperature Range	5% DB	MDBR	7.7	9.3	11.6	12.4	12.0	12.6	12.9	13.1	12.7	10.8	8.5	7.7
		MCDBR	9.8	13.0	16.5	17.6	16.5	17.7	17.5	18.3	16.9	14.0	11.8	10.2
		MCWBR	6.7	8.5	9.8	9.6	8.2	7.6	6.9	7.0	8.0	7.5	7.7	7.0
	5% WB	MCDBR	8.9	11.8	14.0	15.9	14.2	15.2	15.6	15.6	14.3	12.6	10.7	9.0
		MCWBR	6.4	7.9	8.6	8.9	7.6	6.9	6.5	6.6	7.1	7.2	7.2	6.4
Clear-Sky Solar Irradiance	taub	0.305	0.325	0.354	0.384	0.390	0.398	0.393	0.388	0.361	0.353	0.321	0.302	
	taud	2.481	2.422	2.358	2.280	2.314	2.326	2.334	2.360	2.412	2.437	2.482	2.484	
	Ebn at Noon	801	849	871	872	877	869	868	859	853	804	771	762	
	Edh at Noon	67	86	105	124	125	124	121	114	99	84	67	60	
All-Sky Solar Radiation	RadAvg	1.19	2.02	3.25	4.40	5.15	6.00	5.86	5.13	3.97	2.50	1.39	1.01	
	RadStd	0.15	0.29	0.39	0.55	0.49	0.55	0.45	0.43	0.28	0.19	0.15	0.17	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (25 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page